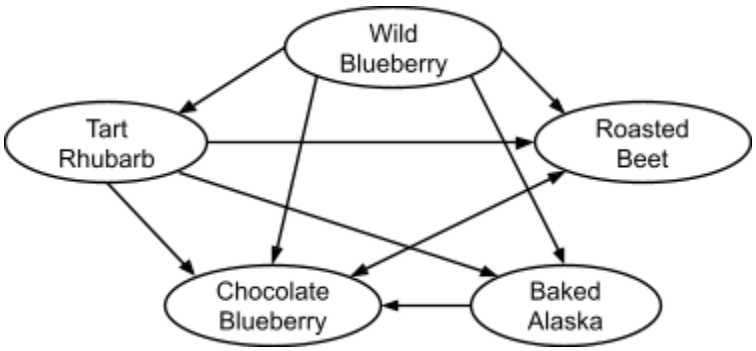
American Computer Science League

2025-2026 ● Contest 4: Short Problems ● Intermediate Division

1. Graph Theory

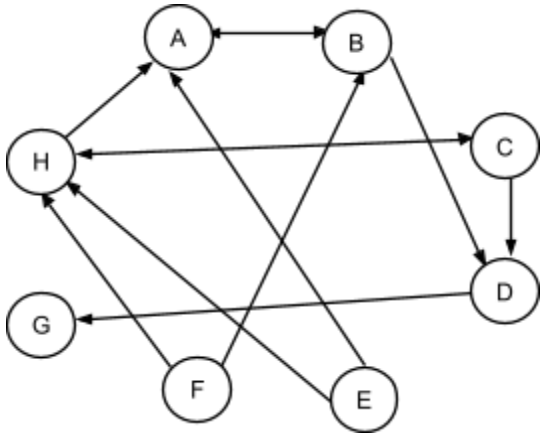
Sally goes to Wild Scoops to buy a 3-scoop ice cream cone. She needs to choose which scoop is on the bottom, middle, and top. The rules of the shop say that the scooper can only be used in a certain order. For example, from the graph below, if you scoop Baked Alaska, then only Chocolate Blueberry can go on top of it. Which of the following is NOT possible from bottom to top?



A. Tart Rhubarb, Chocolate Blueberry, Roasted Beet
B. Baked Alaska, Chocolate Blueberry, Roasted Beet
C. Wild Blueberry, Tart Rhubarb, Baked Alaska
D. Wild Blueberry, Baked Alaska, Chocolate Blueberry
E. Roasted Beet, Tart Rhubarb, Baked Alaska

2. Graph Theory

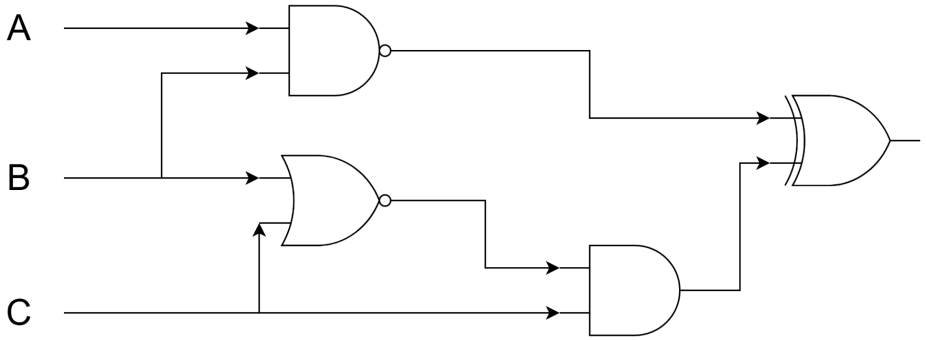
How many vertices do not appear in any cycles of this directed graph?



A. 3
B. 4
C. 5
D. 6
E. 7

3. Digital Electronics

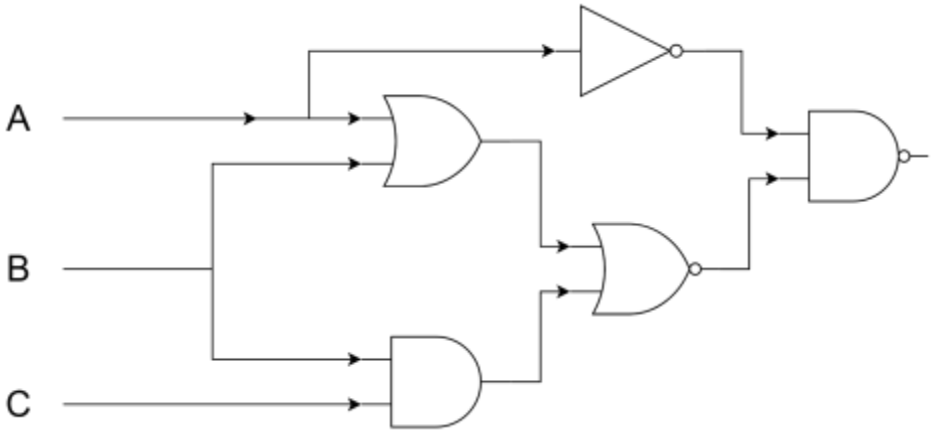
How many ordered triples make the following digital circuit TRUE?



A. 0
B. 2
C. 4
D. 6
E. 8

4. Digital Electronics

Simplify the Boolean algebra expression represented by the following digital circuit:



A. $\overline{A}$
B. ABC
C. $A + BC$
D. $A + B$
E. BC

5. Assembly Language

What is the output when the following program is executed?

```
A       DC      7
B       DC      8
        LOAD    B
        SUB     =10
        MULT    =-1
        STORE   C
        LOAD    B
        MULT    C
        STORE   D
        LOAD    A
        MULT    B
        MULT    =100
        ADD     D
        STORE   ANS
        PRINT   ANS
        END
```

A. 72
B. 1224
C. 3024
D. 4221
E. 5616

6. Assembly Language

What is the output when the following program is executed with these data values: 24, 11, 22, 35, 13, 26, 7, 0, -3?

```
ANS     DC      0
        READ    P
TOP     READ    N
        LOAD    N
        BE      STOP
        LOAD    P
        SUB     N
        BL      CONT
        LOAD    ANS
        ADD     =1
        STORE   ANS
CONT    LOAD    N
        STORE   P
        BU      TOP
STOP    PRINT   ANS
        END
```

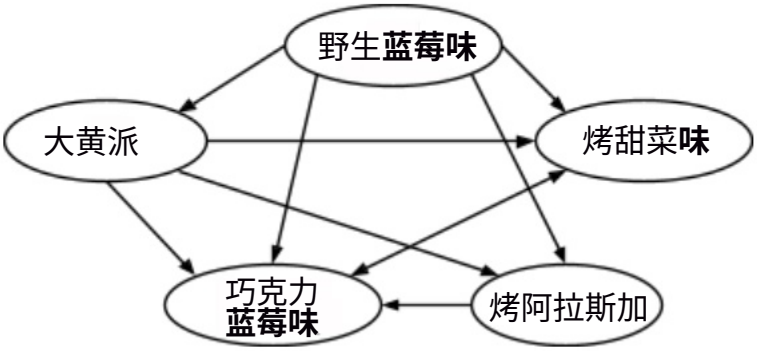
A. 0
B. 1
C. 2
D. 3
E. 4

American Computer Science League

2025-2026 ● Contest 4: Short Problems ● Intermediate Division

1. 图论

Sally去冰淇凌店买一个三球冰淇淋甜筒。她需要选择底部、中部和顶部分别是什么口味。店铺规定，挖球的顺序必须遵循特定规则。例如，根据下图所示，如果先挖了烤阿拉斯加口味，那么只有巧克力蓝莓口味可以放在它上面。请问从下到上，以下哪种组合不可能实现?



A.
大黄派
巧克力蓝莓味
烤甜菜味

B.
烤阿拉斯加
巧克力蓝莓味
烤甜菜味

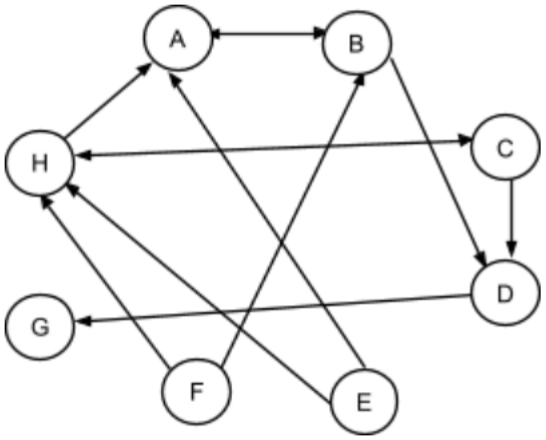
C.
野生蓝莓味
大黄派
烤阿拉斯加

D.
野生蓝莓味
烤阿拉斯加
巧克力蓝莓味

E.
烤甜菜味
大黄派
烤阿拉斯加

2. 图论

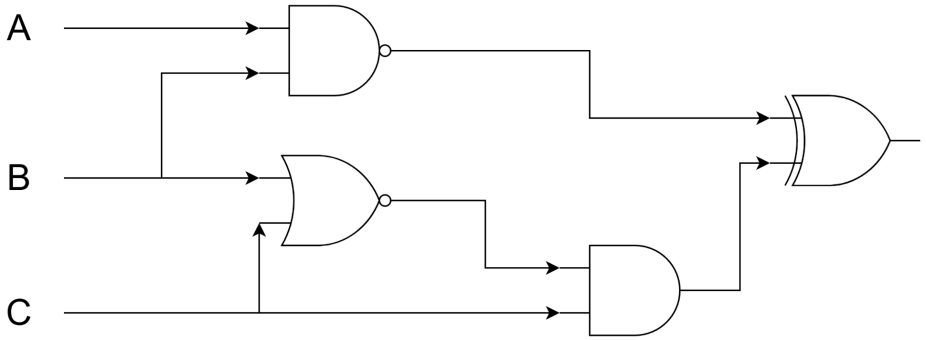
请问在这个有向图中，有多少个顶点不出现在任何环中?



A. 3
B. 4
C. 5
D. 6
E. 7

3. 数字电路

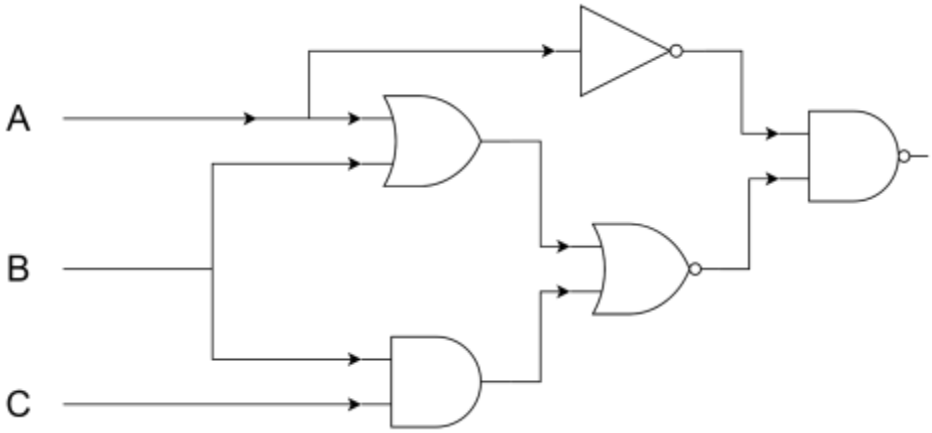
请问有多少个有序三元组可以使得下述数字电路为真？



A. 0
B. 2
C. 4
D. 6
E. 8

4. 数字电路

请化简下图所示数字电路所对应的布尔代数表达式：



A. $\overline{A}$
B. ABC
C. $A + BC$
D. $A + B$
E. BC

5. 汇编语言

请问执行下列程序时会输出什么？

```
A       DC      7
B       DC      8
        LOAD    B
        SUB     =10
        MULT    =-1
        STORE   C
        LOAD    B
        MULT    C
        STORE   D
        LOAD    A
        MULT    B
        MULT    =100
        ADD     D
        STORE   ANS
        PRINT   ANS
        END
```

A. 72
B. 1224
C. 3024
D. 4221
E. 5616

6. 汇编语言

当使用以下数据值: 24, 11, 22, 35, 13, 26, 7, 0, -3 执行**该程序时，输出结果是什么**?

```
ANS     DC      0
        READ    P
TOP     READ    N
        LOAD    N
        BE      STOP
        LOAD    P
        SUB     N
        BL      CONT
        LOAD    ANS
        ADD     =1
        STORE   ANS
CONT    LOAD    N
        STORE   P
        BU      TOP
STOP    PRINT   ANS
        END
```

A. 0
B. 1
C. 2
D. 3
E. 4